

Úloha 12

Zadání

$$y' - xy = -y^3 e^{-x^2}$$

Řešení

$$y' - xy = -y^3 e^{-x^2}$$

$$y' y^{-3} = -e^{-x^2} + xy^{-2}$$

$$\begin{bmatrix} z = y^{-2} \\ z' = -2y^{-3}y' \end{bmatrix}$$

$$\frac{-1}{2}z' - xz = -e^{-x^2}$$

$$z' + 2xz = 2e^{-x^2}$$

$$z'e^{x^2} + 2xe^{x^2}z = 2$$

$$(ze^{x^2})' = 2$$

$$ze^{x^2} = 2x + C$$

$$z = (2x + C)e^{-x^2}$$

$$y = \pm \frac{e^{\frac{x^2}{2}}}{\sqrt{2x + C}}$$

Výsledek

$$y = \frac{e^{\frac{x^2}{2}}}{\sqrt{2x + C}} \quad x \in \left(-\frac{C}{2}, +\infty\right)$$

$$y = -\frac{e^{\frac{x^2}{2}}}{\sqrt{2x + C}} \quad x \in \left(-\frac{C}{2}, +\infty\right)$$

$$y \equiv 0 \quad x \in \mathbb{R}$$